Brain Interacts

WHITEPAPER

The Nudge Enterprise

Using choice architecture to drive software adoption.

The Behavioral Enterprise Series

Brain Interacts

Written for

CIOs · CTOs · Chief Digital Officers · Heads of Change Management · Owners of CRM, ERP and Collaboration Platforms

AT A GLANCE

Four ideas in one page

The problem

Enterprises buy software on the assumption that people will use it once they are told to and trained to. Usage data says otherwise.

The insight

People rarely weigh every option afresh. They follow the path the environment lays out, and whoever designs that path, deliberately or by accident, determines adoption.

The method

The DEFAULT Audit: a seven-part review of the choice environment around any enterprise tool.

The payoff

Higher active use, cleaner data, fewer wasted licenses and less change fatigue, achieved mostly through design changes that are cheap, fast and reversible.

THE CENTRAL POINT

No design is neutral. Every enterprise tool already has a choice architecture. The only question is whether it was designed on purpose.

THE SCENE

Paying for software nobody uses

A company signs a multi-year contract for a new CRM or ERP module. The business case promises a single source of truth, faster forecasting and fewer manual reports. A go-live date is set, an executive email goes out, and every employee completes a 45-minute e-learning module.

Eighteen months later the pipeline in the system and the pipeline in the leadership spreadsheet no longer agree. The best performers keep their own notes. Data is entered in bursts before the quarterly review. The vendor's renewal quote arrives for the full seat count.

This is not an unusual failure; it is the norm. Zylo, a SaaS management vendor, publishes an annual index drawn from its own customer base.

46%	$19.8M	$80.6M	70%
of SaaS licenses unused in a given 30-day period	average annual waste per organization	at enterprises with 10,000+ employees	of contracts renewed despite underuse (2023)

Source: Zylo SaaS Management Index, 2026 edition unless noted.

Earlier editions put the unused share anywhere from about 44% to 53%, so the exact figure moves, but the direction does not. Because these are vendor-sourced figures from a self-selected customer set, they are best read as directional.

Organizations routinely pay for software that employees quietly decline to use, and then renew it.

THE ARGUMENT

Adoption is a problem of design

The usual explanation is "change resistance," and the usual remedy is more communication, more training and, eventually, a mandate. You may also have heard that 70% of change programs fail. That statistic has no solid empirical basis (Hughes, 2011), but the experience behind it is real, and the pattern is consistent.

THE USUAL VIEW

A problem of knowledge and willpower

Mandates and training modules: if people understood the benefits and were told clearly enough, they would comply.

THE BEHAVIORAL VIEW

A problem of design

Behavior is shaped by the environment in which choices are made.

Most enterprise environments were built for the convenience of the system, not the person using it.

The same research that explains why employees default to spreadsheets also shows how to make the new tool the path of least resistance, without removing anyone's freedom to choose.

THE SCIENCE

Why defaults beat directives

STATUS QUO BIAS

We stick with what is already there

The tendency to stick with the current, pre-selected or easiest option, even when alternatives are objectively better (Samuelson & Zeckhauser, 1988). Switching carries costs: effort, uncertainty and the possibility of regret. When the old way still works, inertia usually wins.

CHOICE ARCHITECTURE

The design of the context in which people choose

A term popularized by Thaler and Sunstein (2008). No design is neutral. Someone has to decide what a form shows first, which option is pre-selected, how many clicks a task takes and when a reminder appears. Every enterprise tool already has a choice architecture. The only question is whether it was designed on purpose.

NUDGE

Any aspect of that architecture that predictably changes behavior without forbidding options or significantly changing economic incentives.

Five bodies of evidence are particularly relevant to enterprise technology.

Defaults | **High-stakes defaults** | **Commitment** | **Peers** | **Ease and usefulness**

THE EVIDENCE

37% → 86%

Madrian & Shea, 2001 · Choi, Laibson & Madrian, 2004

Defaults move behavior more than education does

A large U.S. employer switched its 401(k) plan from opt-in to automatic enrollment. Participation among new hires rose by 48 percentage points, from about 37% to 86%. A follow-up analysis of three companies found participation of 86% to 96% under automatic enrollment. Nobody was forced; employees could opt out at any time.

A warning that matters for enterprise design: a substantial share of employees stuck with the default contribution rate and fund, even though few earlier hires had chosen that combination. Defaults are powerful, so they must be chosen with care.

99.98% vs ~12%

Johnson & Goldstein, 2003

Defaults shape outcomes in high-stakes settings

Comparing organ-donation policies across European countries, effective consent was 99.98% in Austria, where citizens are presumed to be donors unless they opt out, and around 12% in neighboring Germany, which requires an explicit opt-in.



99.98%
Austria · opt out

~12%
Germany · opt in

The comparison reports effective consent, not realized donations, and later research shows that the real-world gap is smaller. The lesson about the pull of defaults stands.

THE EVIDENCE, CONTINUED

3.5% → 13.6%

Thaler & Benartzi, 2004

Commitment and timing can be designed

In the Save More Tomorrow program, employees agreed in advance to direct part of future pay raises into retirement savings. Participants who stayed in the program raised average savings rates from 3.5% to 13.6% over 40 months. The trick was moving the decision to a moment when it felt painless and making continuation the default.

~2%

Allcott, 2011

Peers are persuasive

In large-scale field experiments, households received reports comparing their energy use with their neighbors'. Average consumption fell by about 2%. The effect is modest per household but has a near-zero marginal cost, which is what makes it attractive at enterprise scale.

Useful + easy

Davis, 1989 · Venkatesh et al., 2003

Ease and usefulness drive technology acceptance

Perceived usefulness and perceived ease of use predict whether people adopt a technology. The unified model of technology acceptance adds social influence and facilitating conditions.

These map neatly onto nudge design: defaults and ease reduce effort, peers supply social influence, and cues and commitments create the facilitating conditions that turn intent into action.

HONEST LIMITS

Nudges are not magic. DellaVigna and Linos (2022) compared published academic nudge studies with trials run by two large government nudge units, largely finding that published studies are a selective sample.

8.7 pts
Average effect in the academic literature

1.4 pts
Average effect in nudge-unit trials

The practical conclusion is to expect smaller effects than headline studies suggest and to test every intervention in your own environment.

Psychological reactance, the urge to resist perceived pressure (Brehm, 1966), also explains why heavy-handed mandates often provoke the behavior they try to prevent.

Expect smaller effects than the headlines. Test every nudge in your own environment.

DIAGNOSIS

How poor choice architecture shows up

Software adoption failure rarely announces itself as a single event. It appears as a cluster of symptoms that, once you know the behavioral causes, are easier to diagnose.

SYMPTOM IN OPERATIONS	WHAT YOU SEE	LIKELY BEHAVIORAL DRIVER
Parallel spreadsheets	Leaders ask for "the real numbers" in a separate file	Status quo bias; the old path has lower switching cost
Late or incomplete data	Records updated in bursts before reviews; optional fields left blank	Present bias; effort is immediate, benefit is deferred
Training spike, then decay	Logins peak after go-live and fall within weeks	No cue at the moment of need; forgetting
Power-user concentration	A handful of enthusiasts do most of the work	Absence of visible social proof; weak norms
Mandate backlash	Compliance in letter but not spirit; workarounds multiply	Psychological reactance
Renewal inertia	Contracts renewed with high shelfware	Defaults in procurement; nobody owns the utilization number

Employees are making rational, local decisions inside an environment that rewards the old behavior. Asking them to try harder is a weak strategy. Changing the environment is stronger.

THE METHOD

The DEFAULT Audit

A structured review of the choice environment around a single tool or workflow.

Each letter names a design lever, and together they cover how people decide, notice, persist and learn.

DECIDE

- **D** Defaults
- **E** Ease
- **F** Framing

NOTICE

- **A** Attention cues
- **U** Use of peers

PERSIST

- **L** Lock in

T — **LEARN · TEST**
Treat every lever as a hypothesis and measure it

Seven design levers grouped by what they do for the user.

START HERE

Defaults and ease are usually the cheapest and the most powerful levers.

THE LEVERS · DECIDE

D Defaults

Make the desired behavior the pre-selected one.

IN PRACTICE · Provision accounts on day one with role-based views already configured; set the team dashboard as the home screen; turn on the notifications that matter by default; auto-create a record when a customer email arrives and pre-assign an owner; pre-fill the next step on every opportunity. Choose each default for the median user's genuine interest and write down the reasoning, so it can be reviewed.

E Ease

The compliant path should be the shortest one.

IN PRACTICE · Count the clicks and fields behind the ten most common tasks and remove everything that does not feed a decision. Integrate with email and calendar so data enters once. Offer mobile entry for field staff. Adopt a one-minute rule: if a routine update takes longer than a minute, redesign the form, not the person.

F Framing

Describe the tool in terms of what the user gains or avoids losing, not in terms of features or compliance.

IN PRACTICE · Replace "Update the CRM by Friday" with "Logged deals are protected in territory disputes and drop out of the Monday status meeting." People weigh losses more heavily than equivalent gains, so name concrete, honest losses avoided, and give back time wherever the tool removes a task. Never frame with claims that are not true.

THE LEVERS · NOTICE, PERSIST, LEARN

A Attention cues

Prompt at the moment of need, inside the flow of work.

IN PRACTICE · A one-click prompt after a customer meeting ends ("Log this call?"), a helpful empty-state message on a blank dashboard, a progress indicator on an incomplete record. Avoid blanket email reminders and cap prompt frequency; too many cues create alert fatigue and are quickly ignored.

U Use of peers

Make the behavior of colleagues visible, because people follow the norms of those around them.

IN PRACTICE · Show team-level adoption ("9 of 10 people on your team already use the Forecast view"), recruit a champions network drawn from respected peers rather than from IT, and share quick wins in team channels. Prefer team-level signals to individual rankings, which can feel like surveillance and breed resentment.

L Lock in commitments

Let people commit in advance, when the decision feels easy.

IN PRACTICE · Ask teams to agree on one concrete trigger and action before launch ("After every customer call I will log it before opening email"). Invite visible team pledges. Where appropriate, use pre-commitment in the Save More Tomorrow style, such as agreeing now that new reports will be built only in the new tool from a set date.

T Test

Treat every nudge as a hypothesis.

IN PRACTICE · Run staggered or randomized rollouts across teams or regions, measure behavior rather than satisfaction, and retire nudges that do not work. Because real-world effects tend to be smaller than published ones (DellaVigna & Linos, 2022), the ability to test at your own scale is what keeps the program honest.

METHOD

Running the audit

MAP

Map the moments of choice

Walk the user journey from access request through first use, daily use and renewal. At each moment, write down what the user decides and what the environment currently suggests.

INVENTORY

Inventory defaults and friction

Record the current default, the number of steps and the cue, if any. Watch people who use the old workaround; the workaround is a map of where the new tool is failing them.

SCORE

Score and prioritize

Rate each lever from absent to strong at each moment, then rank improvements by expected impact against effort. Start with defaults and ease.

PILOT

Pilot and iterate

Pick two or three changes, run them against a comparison group and decide based on observed behavior.

A SIMPLE WORKING TEMPLATE

MOMENT OF CHOICE	CURRENT DEFAULT	DESIRED BEHAVIOR	NUDGE (LEVER)	OWNER	METRIC
Opportunity created	Blank record, owner unset	Owner and next step set at creation	Pre-fill from email (D, E)	Sales Ops	% records with owner within a day
After a customer call	No prompt	Call logged same day	One-click post-meeting prompt (A)	CRM admin	Median logging delay
Weekly forecast review	Spreadsheet accepted	Forecast read from system	Review runs only from system view (D, F)	VP Sales	Spreadsheet-sourced reports
New-hire onboarding	Access on request	Configured account on day one	Role-based provisioning (D)	IT / HR	Time to first value

CASE ILLUSTRATION

Meridian Components

A composite illustration, not a report on a named organization. All figures are illustrative.

A hypothetical industrial manufacturer with 4,200 employees replaced a legacy CRM for roughly 900 sales and service staff. The first rollout followed the traditional playbook: an executive announcement, mandatory training and a go-live deadline. Leadership still ran forecast reviews from regional spreadsheets. Six months later:

71%	48%	39%
logged in at least once a week	genuinely active (3+ records updated weekly)	of opportunity changes entered within two business days

INSTEAD OF A SECOND TRAINING PUSH, THE CIO AND HEAD OF SALES OPERATIONS RAN A DEFAULT AUDIT

DEFAULTS · EASE	Opportunities were auto-created from inbound email with the owner pre-assigned, and the five most-used fields were pre-filled. Seven rarely used required fields were dropped.
ATTENTION	A one-click prompt after each calendar meeting with an external contact replaced weekly reminder emails.
FRAMING	Communications shifted from "compliance" to "time returned," because the weekly status meeting was cancelled once the system view was up to date.
PEERS	Each team saw a team-level completeness score, and champions were named in every region.
TEST	Six regions moved to the new setup in staggered waves, with the first two serving as a comparison group.

MERIDIAN COMPONENTS, CONTINUED

TWO THINGS DID NOT WORK

- ✕ Reminder emails produced no measurable change.
- ✕ An early individual leaderboard drew complaints about surveillance, so it was removed after three weeks.

Both findings came from testing, not from opinion.

ILLUSTRATIVE RESULTS AFTER SIX MONTHS

Active users (3+ records updated weekly) **48% → 81%**	Opportunity updates within two business days **39% → 74%**
Forecast variance versus actuals (average) **±22% → ±11%**	Seats reclaimed at renewal **9% of licenses**

No single nudge produced the gain. The combined redesign made the new tool the easiest way to do the job, and the team knew which parts worked because it measured them.

MEASUREMENT

What to track, and what good looks like

Measure behavior, not attendance. Logins and training completions are easy to report and poor indicators of value. Define active use as a specific action that creates business value, and track depth of use, data quality and speed alongside breadth. Baselines and targets are illustrative.

KPI	DEFINITION	ILLUSTRATIVE BASELINE	TARGET (6 MONTHS)
Weekly active users	Users performing the core value action at least N times a week	45–55% of licensed users	75–85%
Feature depth	Share of active users using at least three core features	20–30%	50–60%
Data completeness	Required and decision-critical fields populated	60–70%	90%+
Update timeliness	Records updated within two business days of the event	35–45%	70–80%
Time to first value	Days from account creation to first completed core task	14–21 days	3–5 days
License utilization	Active seats divided by paid seats	50–55%	80–90%
Workaround usage	Reports or decisions still sourced from spreadsheets	High; unmeasured	Down by 50% or more
Nudge opt-out rate Guardrail	Share of users turning off prompts	—	Below 15%; investigate if higher
Friction score Guardrail	Average effort rating for five core tasks	Measure at start	Improve by one point or more

TRACK THE GUARDRAILS

A rising opt-out rate or a worsening effort score means the nudges are becoming noise or pressure.

KEEP A COMPARISON GROUP

Without staggered rollouts, you cannot tell a nudge effect from a seasonal pattern or a vendor release.

GUARDRAILS

Where nudging goes wrong

A nudge cannot fix a bad tool. If the software does not help people do their jobs, no default or prompt will produce lasting use. Davis's research on usefulness and ease of use points the same way. Listen to the workarounds; they are product feedback.

Nudge overload. Stacking prompts, banners and badges produces alert fatigue. Budget the number of cues a user sees each day and remove those that do not earn their place.

Defaults that serve the organization at the user's expense. Defaults that quietly opt employees into data sharing, extra notifications outside working hours or extended monitoring cross the line from helpful design to manipulation. Madrian and Shea's finding that many employees kept a default they would not have chosen is a reminder that defaults carry responsibility.

Vanity metrics and Goodhart's law. Once a measure becomes a target, people optimize it. If you reward logins, you get logins. Prefer outcome-linked metrics and review them regularly.

Effect decay and novelty. Prompts lose force over time. Rotate and refresh them, and pair nudges with the habit-building methods in Atomic Habits at Scale.

AN ETHICS TEST BEFORE LAUNCHING ANY NUDGE

TRANSPARENCY
Would the employee endorse the design if they understood it?

FREEDOM
Can they opt out easily?

BENEFIT
Does it serve their interest as well as the company's?

If the answer to any of them is no, redesign.

LEGAL AND COMPLIANCE NOTE

Adoption programs often rely on usage analytics. In the European Union and many other jurisdictions, collecting and analyzing employee usage data can require a lawful basis, a data protection impact assessment, transparency to employees and, in some countries, consultation with works councils or employee representatives. Individual-level monitoring raises higher risks than team-level aggregates. Involve legal and data protection teams before launching analytics, and favor aggregated, team-level reporting by default. This paper is not legal advice.

THE PLAN

The first 90 days

DAYS 1–30

Diagnose

- Choose one high-value tool or workflow with a clear owner and a measurable business outcome.
- Define active use as a behavior and capture a baseline for the KPIs.
- Interview and observe five to eight users, including people who rely on workarounds.
- Run the DEFAULT Audit and shortlist five changes, favoring defaults and ease.
- Secure an executive sponsor and complete a legal and data-protection review of the analytics you plan to use.

DAYS 31–60

Pilot

- Implement the top three to five changes for a pilot group, with a comparable group as control.
- Launch a champions network of respected peers and brief them on the new defaults.
- Review behavior data weekly; remove any nudge that shows no effect or draws complaints.
- Gather qualitative feedback and check the guardrail metrics.

DAYS 61–90

Scale and embed

- Roll out the changes that worked to the remaining teams; retire the ones that did not.
- Build the winning defaults into onboarding and role-based provisioning so they persist.
- Report results to the sponsor in business terms: time saved, data quality, license recovery, forecast accuracy.
- Feed utilization data into the next renewal negotiation and set a quarterly review of the choice architecture.

CALL TO ACTION

Pick one tool your organization is already paying for and run the audit on it this quarter. You do not need a new platform, a bigger training budget or a reorganization. You need to look at the tool the way your newest employee does, ask what the environment is nudging them to do, and change it on purpose.

CONTINUE READING

Connections to the series

Frictionless Compliance	Security and regulatory protocols are choice architecture too; the same ease and defaults principles apply.
Atomic Habits at Scale	Nudges start behavior; habit loops sustain it after the novelty fades.
The ROI of Psychological Safety	Employees must feel safe saying that a tool makes their work harder, or the best diagnostic information never reaches you.
The Sunk Cost Trap	If testing shows a tool cannot reach adoption, the organization needs a disciplined way to stop investing in it.
Beyond the Bonus	Autonomy and mastery explain why people embrace tools that make them better at their jobs and resist tools that merely monitor them.

Brain Interacts

ABOUT BRAIN INTERACTS

Brain Interacts is a team analytics and behavioral analysis company. We build Corporate Brain and work with organisations on culture studies, root-cause analysis, and behavior-led operating design.

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SOURCES

References

Allcott, H. (2011). Social norms and energy conservation. *Journal of Public Economics*, 95(9–10), 1082–1095.

Brehm, J. W. (1966). *A Theory of Psychological Reactance*. Academic Press.

Choi, J. J., Laibson, D., & Madrian, B. C. (2004). Plan design and 401(k) savings outcomes. NBER Working Paper 10486.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.

DellaVigna, S., & Linos, E. (2022). RCTs to scale: Comprehensive evidence from two nudge units. *Econometrica*, 90(1), 81–116.

Hughes, M. (2011). Do 70 per cent of all organizational change initiatives really fail? *Journal of Change Management*, 11(4), 451–464.

Johnson, E. J., & Goldstein, D. (2003). Do defaults save lives? *Science*, 302(5649), 1338–1339.

Madrian, B. C., & Shea, D. F. (2001). The power of suggestion: Inertia in 401(k) participation and savings behavior. *Quarterly Journal of Economics*, 116(4), 1149–1187.

Samuelson, W., & Zeckhauser, R. (1988). Status quo bias in decision making. *Journal of Risk and Uncertainty*, 1(1), 7–59.

Thaler, R. H., & Benartzi, S. (2004). Save More Tomorrow: Using behavioral economics to increase employee saving. *Journal of Political Economy*, 112(S1), S164–S187.

Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: Improving Decisions About Health, Wealth, and Happiness*. Yale University Press.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.

Zylo. (2026). SaaS Management Index (and 2023 and 2025 editions). Vendor-published benchmark data.